



White Band Associates

WHITE BAND ASSOCIATES

Cyber Security

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Networking Task 02

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Week 1 - Networking

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INTRODUCTION

Networking is the foundation of modern communication systems and plays a vital role in connecting devices and enabling data exchange over the Internet. Understanding network devices, IP addressing, DNS services, and network communication processes is essential for building a strong foundation in Cyber Security.

This task focuses on exploring common network devices, classifying IP addresses, analyzing network configuration, understanding how data travels across a network, and performing practical networking commands using Windows and Kali Linux. The activities performed in this task provide hands-on experience with fundamental networking concepts and tools used in real-world environments.

OBJECTIVE

- To understand the purpose and functioning of common network devices.
- To learn the difference between Public and Private IP addresses.
- To identify and analyze network configuration details on a local device.
- To understand the role of DNS and network communication flow.
- To perform practical networking commands in Windows and Kali Linux.
- To develop a strong foundation in networking concepts essential for Cyber Security.

NETWORK DEVICES RESEARCH

Network devices are hardware components that help in communication, data transfer, and connectivity within a network. They play an important role in connecting devices, managing network traffic, providing Internet access, and ensuring secure communication between systems.

Router

Purpose : Connects different networks and provides Internet access to devices.

How It Works : Uses IP addresses to forward data packets to their correct destination.

Real-World Usage : Used in homes and offices to connect devices to the Internet.

Switch

Purpose : Connects multiple devices within the same network.

How It Works : Uses MAC addresses to send data only to the intended device.

Real-World Usage : Commonly used in offices, schools, and organizations.

Hub

Purpose : Connects multiple devices in a network.

How It Works : Sends received data to all connected devices without checking the destination.

Real-World Usage : Previously used in small networks, now mostly replaced by switches.

Access Point

Purpose : Provides wireless connectivity to network devices.

How It Works : Connects wireless devices to a wired network using Wi-Fi signals.

Real-World Usage : Found in homes, colleges, offices, and public Wi-Fi areas.

Firewall

Purpose : Protects a network from unauthorized access and threats.

How It Works : Monitors and filters network traffic based on security rules.

Real-World Usage : Used in personal computers, organizations, and enterprise networks.

Modem

Purpose : Connects a network to the Internet Service Provider (ISP).

How It Works : Converts digital data into signals that can travel through communication lines and vice versa.

Real-World Usage : Used in homes and offices to access the Internet.

IP ADDRESS CLASSIFICATION

IP addresses are unique numerical identifiers assigned to devices connected to a network. They are classified as Public or Private based on their accessibility and usage. Private IP addresses are used within local networks, while Public IP addresses are accessible over the Internet.

IP Address	Classification	Reason
192.168.1.10	Private	This IP address belongs to the 192.168.0.0 – 192.168.255.255 private address range reserved for use within local networks and is not directly accessible from the Internet.
10.0.0.5	Private	This IP address falls within the 10.0.0.0 – 10.255.255.255 private address range, which is commonly used in organizations and local area networks.
172.16.5.20	Private	This IP address belongs to the private range 172.16.0.0 – 172.31.255.255, which is reserved for internal network communication.

8.8.8.8	Public	This is a publicly accessible IP address provided by Google and can be reached over the Internet from anywhere in the world.
1.1.1.1	Public	This is a publicly accessible IP address provided by Cloudflare and is available for global Internet users.
192.168.100.1	Private	This IP address belongs to the 192.168.0.0 – 192.168.255.255 private address range and is commonly used as a local router or gateway address within private networks.

UNDERSTANDING YOUR NETWORK

Understanding network configuration helps identify how a device communicates within a network and accesses Internet resources. Important network parameters such as IP Address, Default Gateway, and DNS Server play a key role in establishing connectivity and enabling communication between devices.

Network Information :

Parameter	Value
Hostname	LAPTOP-UPABTK5H
IPv4 Address	10.180.51.70
Default Gateway	10.180.51.197
DNS Server	10.180.51.197

Q1. Which IP range does your device belong to?

My device has the IPv4 address 10.180.51.70, which belongs to the 10.0.0.0 – 10.255.255.255 IP address range.

Q2. Is the IP Address Public or Private?

The IP address 10.180.51.70 is a Private IP Address because it falls within the private address range 10.0.0.0 – 10.255.255.255, which is reserved for internal network communication.

Q3. What role does the router play in your network?

A router connects the local network to other networks, including the Internet. It forwards data packets between devices and ensures that network traffic reaches the correct destination.

Q4. What would happen if the DNS server stopped working?

If the DNS server stopped working, domain names such as google.com would not be translated into IP addresses. As a result, users would have difficulty accessing websites using their domain names even if Internet connectivity was available.

Figure 1 : Network Configuration Details – Part 1

```
hernet adapter VMware Network Adapter VMnet8:
Connection-specific DNS Suffix . : 
Description . . . . . : VMware Virtual Ethernet Adapter for VMnet8
Physical Address. . . . . : 
DHCP Enabled. . . . . : 
Autoconfiguration Enabled . . . . : 
Link-local IPv6 Address . . . . . : 
IPv4 Address. . . . . : 
Subnet Mask . . . . . : 
Lease Obtained. . . . . : 
Lease Expires . . . . . : 
Default Gateway . . . . . : 
DHCP Server . . . . . : 
DHCPv6 IAID . . . . . : 
DHCPv6 Client DUID. . . . . : 
Primary WINS Server . . . . . : 
NetBIOS over Tcpip. . . . . : 

Wireless LAN adapter Wi-Fi:
Connection-specific DNS Suffix . : 
Description . . . . . : 
Physical Address. . . . . : 
DHCP Enabled. . . . . : 
Autoconfiguration Enabled . . . . : 
IPv6 Address. . . . . : 
Temporary IPv6 Address. . . . . : 
Link-local IPv6 Address . . . . . : 
IPv4 Address. . . . . : 
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : 
Lease Expires . . . . . : 
Default Gateway . . . . . : fe80::5cf9:1aff:fe01:6f1e%10
                               10.180.51.197
DHCP Server . . . . . : 
DHCPv6 IAID . . . . . : 
DHCPv6 Client DUID. . . . . : 
DNS Servers . . . . . : 10.180.51.197
                               2409:40c2:6423:6420::b7
NetBIOS over Tcpip. . . . . :
```

Figure 2: Network Configuration Details – Part 2

```
C:\Users\bhakt>ipconfig /all
```

Windows IP Configuration

```
Host Name . . . . . : LAPTOP-UPABTK5H
Primary Dns Suffix . . . . . :
Node Type . . . . . :
IP Routing Enabled. . . . . :
WINS Proxy Enabled. . . . . :
```

Wireless LAN adapter Local Area Connection* 1:

```
Media State . . . . . :
Connection-specific DNS Suffix . :
Description . . . . . :
Physical Address. . . . . :
DHCP Enabled. . . . . :
Autoconfiguration Enabled . . . . :
```

Wireless LAN adapter Local Area Connection* 2:

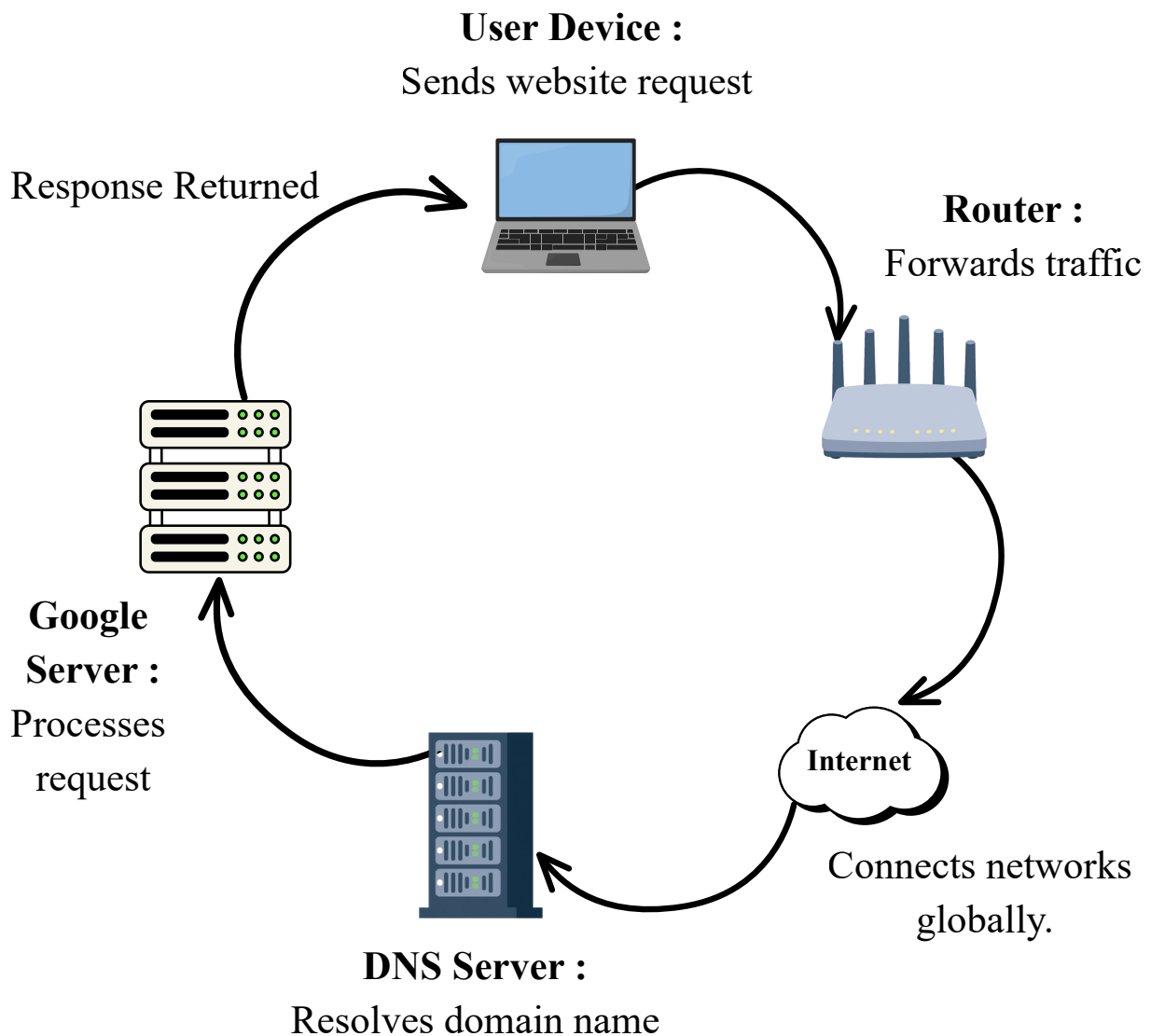
```
Media State . . . . . :
Connection-specific DNS Suffix . :
Description . . . . . :
Physical Address. . . . . :
DHCP Enabled. . . . . :
Autoconfiguration Enabled . . . . :
```

Ethernet adapter VMware Network Adapter VMnet1:

```
Connection-specific DNS Suffix . :
Description . . . . . :
Physical Address. . . . . :
DHCP Enabled. . . . . :
Autoconfiguration Enabled . . . . :
Link-local IPv6 Address . . . . . :
IPv4 Address. . . . . : 192.168.27.1(Preferred)
Subnet Mask . . . . . :
Lease Obtained. . . . . :
Lease Expires . . . . . :
Default Gateway . . . . . :
DHCP Server . . . . . :
DHCPv6 IAID . . . . . :
DHCPv6 Client DUID. . . . . :
NetBIOS over Tcpip. . . . . :
```

NETWORK COMMUNICATION FLOW

Figure 3 : Network Communication Diagram



Network communication flow describes the process through which data travels from a user's device to a destination server and returns with the requested information. This process involves several network components, including the router, DNS server, and web server, working together to establish successful communication.

Step 1 : User Request

The user enters a website address such as google.com in a web browser and sends a request to access the website.

Step 2 : Router Processing

The router receives the request from the user's device and forwards it to the appropriate network destination.

Step 3 : DNS Resolution

The DNS server translates the domain name (google.com) into its corresponding IP address so that the destination server can be located.

Step 4 : Communication with Google Server

Using the IP address obtained from DNS, the request is sent to Google's server through the Internet.

Step 5 : Response Returned

The Google server processes the request and sends the requested webpage data back through the network to the user's device.

PRACTICAL COMMAND EXERCISE

This section demonstrates the use of basic networking commands in Windows and Kali Linux to view network configuration, resolve domain names, and verify network connectivity.

Windows Commands

ipconfig /all

Displays detailed network configuration information of the system.

Figure 4 : Output of ipconfig /all

```
C:\Users\bhakt>ipconfig /all

Windows IP Configuration

Host Name . . . . . : LAPTOP-UPABTK5H
Primary Dns Suffix . . . . . :
Node Type . . . . . :
IP Routing Enabled. . . . . :
WINS Proxy Enabled. . . . . :

Wireless LAN adapter Local Area Connection* 1:

Media State . . . . . :
Connection-specific DNS Suffix . :
Description . . . . . :
Physical Address. . . . . :
DHCP Enabled. . . . . :
Autoconfiguration Enabled . . . . :

Wireless LAN adapter Local Area Connection* 2:

Media State . . . . . :
Connection-specific DNS Suffix . :
Description . . . . . :
Physical Address. . . . . :
DHCP Enabled. . . . . :
Autoconfiguration Enabled . . . . :

Ethernet adapter VMware Network Adapter VMnet1:

Connection-specific DNS Suffix . :
Description . . . . . :
Physical Address. . . . . :
DHCP Enabled. . . . . :
Autoconfiguration Enabled . . . . :
Link-local IPv6 Address . . . . . :
IPv4 Address. . . . . : 192.168.27.1(Preferred)
Subnet Mask . . . . . :
Lease Obtained. . . . . :
Lease Expires . . . . . :
Default Gateway . . . . . :
DHCP Server . . . . . :
DHCPv6 IAID . . . . . :
DHCPv6 Client DUID. . . . . :
NetBIOS over Tcpip. . . . . :
```

nslookup google.com

Retrieves the IP address associated with the domain name google.com.

Observation : The DNS server successfully resolved the domain name and returned the corresponding IP address.

Figure 5 : Output of nslookup google.com

```
C:\Users\bhakt>nslookup google.com
Server:    UnKnown
Address:   10.180.51.197

Non-authoritative answer:
Name:      google.com
Addresses: 2404:6800:4009:804::200e
           142.250.143.113
           142.250.143.139
           142.250.143.138
           142.250.143.101
           142.250.143.100
           142.250.143.102
```

ping google.com

Tests connectivity between the local device and Google's server.

Observation : The ping request was successful, indicating proper network connectivity.

Figure 6 : Output of ping google.com

```
C:\Users\bhakt>ping google.com

Pinging google.com [2404:6800:4009:804::200e] with 32 bytes of data:
Reply from 2404:6800:4009:804::200e: time=53ms
Reply from 2404:6800:4009:804::200e: time=85ms
Reply from 2404:6800:4009:804::200e: time=54ms
Reply from 2404:6800:4009:804::200e: time=62ms

Ping statistics for 2404:6800:4009:804::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 53ms, Maximum = 85ms, Average = 63ms
```

Kali Linux Commands

Kali Linux Network Verification

To view the network interface configuration and verify network connectivity from the Kali Linux system.

Observation : The `ip addr` command displayed the active network interface and assigned IP address, while the `ping google.com` command successfully received replies from Google's server, confirming network connectivity.

Figure 7 : Output of ip addr and ping google.com

A screenshot of a Kali Linux terminal window. The terminal shows the output of the 'ip addr' command, which displays details for the 'eth0' interface, including its state as 'UP' and various network parameters. Below this, the 'ping google.com' command is executed, showing two successful ping requests with their respective IP addresses, TTL values, and response times. The terminal also displays 'ping statistics' at the bottom, indicating 2 packets transmitted with 0% loss and a total time of 1532ms.

```
(kali@kali)~$ ip addr
1:
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000

(kali@kali)~$ ping google.com
PING google.com (2404:6800:4002:82a::200e) 56 data bytes
64 bytes from tzdelb-ay-in-x0e.1e100.net (2404:6800:4002:82a::200e): icmp_seq=1 ttl=116 time=187 ms
64 bytes from tzdelb-ay-in-x0e.1e100.net (2404:6800:4002:82a::200e): icmp_seq=2 ttl=116 time=95.7 ms
^C
-- google.com ping statistics --
2 packets transmitted, 2 received, 0% packet loss, time 1532ms
rtt min/avg/max/mdev = 95.666/141.486/187.306/45.820 ms
```

nslookup google.com

The `nslookup` command is used to query a Domain Name System (DNS) server and obtain the IP address associated with a domain name. It helps verify whether DNS resolution is functioning correctly on the system.

Observation : The command successfully contacted the configured DNS server and resolved the domain name `google.com` into its corresponding IP address. This confirms that DNS services are working properly and the system can translate domain names into IP addresses required for network communication.

Figure 8 : Output of nslookup google.com

```
C:\Users\bhakt>nslookup google.com
Server:   UnKnown
Address:  10.180.51.197

Non-authoritative answer:
Name:     google.com
Addresses: 2404:6800:4009:804::200e
          142.250.143.113
          142.250.143.139
          142.250.143.138
          142.250.143.101
          142.250.143.100
          142.250.143.102
```

Questions and Answers

Q1. What IP address did DNS return for Google?

The DNS server returned the IP address 142.250.206.110 for google.com.

Q2. Was the ping successful?

Yes, the ping was successful and responses were received from the destination server.

Q3. Why is DNS important before communication begins?

DNS translates human-readable domain names into IP addresses, allowing devices to locate and communicate with the correct server on the network.

LEARNING OUTCOMES

- Gained an understanding of common network devices and their functions in a network environment.
- Learned the difference between Public and Private IP addresses and their practical usage.
- Identified and analyzed important network configuration details such as IP Address, Default Gateway, and DNS Server.
- Understood the role of DNS in resolving domain names into IP addresses.
- Learned how data travels through a network using the network communication flow process.
- Performed practical networking commands in both Windows and Kali Linux environments.
- Developed foundational networking knowledge essential for Cyber Security.

CONCLUSION

This task provided valuable knowledge about networking fundamentals, network devices, IP addressing, DNS functionality, and network communication processes. Through both theoretical research and practical command-line exercises, I gained hands-on experience in analyzing network configurations and verifying connectivity. The concepts learned in this task have strengthened my understanding of networking and will serve as a foundation for further learning in Cyber Security.